CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Eight Semester of B. Tech. Examination (EE)

April 2013

EE409 Digital Signal Processing

Date: 25/04/2013 **Time: 10.00 am to 01.00 pm** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q-1 (a) 1. Draw & Explain the operation & application of Barrel shifter used in DSP processor. **5**

2. Explain Pipelining. **2**

OR

(a) 1. Discuss advantages of Digital signal processing over analog signal processing. **(discuss any four in detail)** **7**

Q-2 (a) 1. Determine & give proof for followings **4**

1. $Y(n) = n.x^2(n)$ (Time variant or in-variant?)
2. $Y(n) = 5*y(n-1) + x(n) + 6*y(n+2)$ (Causal or Non causal?)

2. Determine the signal x(n) whose Z transforms is given by $X(Z) = \log(1+az^{-1})$ where $z>a$ **5**

3. Find the Z transform of $x(n) = \cos(w_0 n)$ **5**

OR

(a) 1. Explain how you would determine the Linearity of given system? Also give examples of Linear & Non Linear systems. **4**

2. Derive relationship between Z Transform & DFT. **5**

3. Determine the Z transform of signal $x(n) = (1/2)^n u(n)$ **5**

Q-3 (a) 1. Find the circular convolution of two finite duration sequence **4**

$x_1(n) = \{1, -1, -2, 3, -1\}$

$x_2(n) = \{1, 2, 3\}$

2. Determine the Z transform & ROC of $x(n) = [3(2^n) - 4(3^n)]u(n)$ **5**

3. Give proof of following Z Transform properties. **5**

1. Linearity
2. Time Shifting.

OR

(a) 1. Write step by step procedure to find out Linear or Circular convolution **4**

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2. Give proof of following Z Transform properties. 5

1. Convolution theorem.
2. Time Reversal.

3. Find Z domain equation for following differential questions describing the systems. Also find out the transfer function 5

1. y (n) = 6 y (n-1) + 2 y(n-2) + 2 x (n-1)
2. y (n) = 2 y (n-1) + 3 x (n-1)

OR

3. Evaluate the Linear Convolution of two sequences. 5

$h(n) = (0.5)^n\, u(n)$

$x(n) = 3^n\, u(-n)$

SECTION – II

Q-4 (a) 1. Explain concept of frequency wrapping. 2

2. Draw forms of realization for IIR filter from Its equation. **(attempt any 2 forms)** 5

OR

(a) 1. Write a short note on Filter characteristics of 'various kinds of filter' 7

Q-5 (a) 1. Write down the steps necessary to design analog butterworth filter. 6

2. For Decimation in time algorithm (DIT). Draw signal flow graph 8 point Sequence. Also state necessary equations. 8

OR

(a) 1. Clearly draw the Frequency response characteristic of Low pass Butterworth & Chebyshev filter & discuss the difference. 6

2. Compute the FFT of given sequence using DIT algorithm. Also draw signal flow graph. x(n) = { 2, 3, 10, 8, 7, 9, 7, 6} 8

Q-6 (a) 1. State difference between IIR & FIR filters.**(Mention any six)** 6

2. Discuss Bi-linear transformation technique to design digital IIR filter. 8

OR

1. Using Pole mapping concept, Prove that IIR filter design by 'Approximation of derivatives' results in to a stable filter. 6

2. Realize the digital filter (also draw the figure) using the difference equation (approximation by derivatives) or impulse invariant or Bilinear transformation design method to simulate the analog filter whose transfer function is 8

$$H(s) = \frac{s}{s^2 + 400s + (2 \times 10^5)}$$

Use sampling f_s to be IKHz.